THRIFTKART / SEMESTER III

UML Package

Diagram blueprint + PlantUML source for consistent drawing

Diagrams	Use Case • Class • Sequence • Activity • State
Primary scenario	List a Book and Complete a Sale
State model	Listing lifecycle
Design principle	High cohesion / loose coupling
Drawing tool	Lucidchart / Draw.io / PlantUML
Consistency source	ThriftKart SRS v1.0

1. Use Case Diagram Blueprint

Actor	Use cases
Student Buyer	Register/Login; Find a Book; View Listing; Start Chat; Request Purchase; Complete Sale; View Transaction; Report Listing/User
Student Seller	Register/Login; Manage Profile; Create Listing; Submit for Moderation; Respond in Chat; Complete Sale; View Transaction
Admin Moderator	Review Listing; Approve/Reject Listing; Review Reports; Resolve Report
Core system services	Validate input; persist data; enforce reservation state; record audit evidence

Recommended relationships: “Submit Listing” includes “Validate Listing”; “Review Listing” includes “Record Moderation Decision”; “Request Purchase” includes “Check Availability”; “Complete Sale” includes “Record Payment Status” and “Record Handover”.

2. Class Diagram Blueprint

Class	Core attributes	Core operations
User	user_id, email, name, year, role, status	register(), authenticate(), updateProfile()
Category	category_id, name, slug	listCategories()
Listing	listing_id, seller_id, category_id, title, price, condition, status	validate(), submitForModeration(), reserve(), markSold()
ListingImage	image_id, listing_id, image_url, sort_order	addImage(), removeImage()
Conversation	conversation_id, listing_id, buyer_id, seller_id, status	start(), close()
Message	message_id, conversation_id, sender_id, body, sent_at	send(), markRead()
Transaction	transaction_id, listing_id, buyer_id, seller_id, agreed_price, status	request(), confirmPayment(), scheduleHandover(), complete()
PaymentRecord	payment_id, transaction_id, amount, method, status	record(), confirm()
ModerationReview	review_id, listing_id, moderator_id, decision, reason	approve(), reject()
Report	report_id, reporter_id, listing_id, reported_user_id, reason, status	open(), resolve()

3. Relationship / Multiplicity Sheet

Relationship	Interpretation
User 1 — 0..* Listing	One seller can create many listings; each listing has one seller.
Category 1 — 0..* Listing	One category groups many listings; each listing uses one category.
Listing 1 — 0..* ListingImage	A listing may have multiple images.
Listing 1 — 0..* Conversation	A listing can generate conversations with different buyers.
Conversation 1 — 0..* Message	A conversation contains messages.
Listing 1 — 0..1 active Transaction	A listing may be sold once; historical transaction record is retained.
Transaction 1 — 0..1 PaymentRecord	The MVP may have zero or one payment evidence record.
Listing 1 — 0..* ModerationReview	Keep review history if a listing is re-reviewed.
User 1 — 0..* Report	A user can submit multiple reports and can be the reported user.

4. Sequence Diagram — “List a Book and Complete a Sale”

Step	From	To	Action
1	Seller	Web UI	Open Sell page and enter textbook data.
2	Web UI	API	POST listing payload.
3	API	Database	Validate fields and save Draft.
4	Seller	Web UI	Submit for moderation.
5	Web UI	API	PUT listing status = Pending Moderation.
6	API	Admin	Expose listing in moderation queue.
7	Admin	API	Approve listing and store reason/timestamp.
8	API	Database	Set listing = Published.
9	Buyer	Web UI	Search and open listing.
10	Buyer	API	Request Purchase.
11	API	Database	Verify available; create Transaction and reserve Listing.
12	Buyer ↔ Seller	Chat module	Coordinate price/handover details.
13	Buyer/Seller	Transaction	Record payment status and handover details.
14	Transaction	Database	Mark transaction Completed and listing Sold.

5. Activity Diagram — Suggested Flow

Node	Flow
Start	Student opens ThriftKart
Decision	Buyer or Seller?
Seller path	Create listing → Validate → Submit → Pending Moderation → Admin decision → Published / Rejected
Buyer path	Search → Filter → View listing → Request purchase → Reservation created → Chat/coordinate → Payment status → Handover
Converge	Transaction completed → Listing Sold
End	Show completion state

6. State Diagram — Listing

State	Meaning
Draft	Seller is editing; not visible publicly.
Pending Moderation	Submitted; waiting for admin review.
Published	Approved and discoverable.
Reserved	Buyer has an active purchase request/hold.
Sold	Transaction and handover completed.
Rejected	Moderator declined; seller may correct and resubmit.
Archived	Listing removed from active marketplace by seller/admin.

Transitions: Draft → Pending Moderation; Pending Moderation → Published or Rejected; Rejected → Draft; Published → Reserved or Archived; Reserved → Sold or Published (on release/expiry); Sold → Archived.

7. Cohesion & Coupling Justification

Module	Owns	Does not own
Listing / Marketplace	Book metadata, search and listing state	Payment processing internals
Moderation	Review decision and audit record	Buyer transaction logic
Conversation	Listing-scoped buyer/seller messages	Reservation or payment state
Transaction	Purchase request, reservation, handover and completion	Book discovery/search
Payment	Payment method/status evidence	Listing publication state

Design rule

Reference other modules by stable IDs and service/API boundaries. Do not embed payment, chat or moderation logic inside the Listing entity. This keeps each class/module focused and makes the system easier to test and change.

8. PlantUML Source

The companion .puml file in the deliverable pack contains five diagrams using the same actors, classes, relationships, states and scenario described here. Paste each @startuml block into PlantUML or a compatible renderer, or use the text above as the exact Lucidchart/Draw.io drawing guide.